



MULTI-HORIZON FORECASTING OF CARDIOVASCULAR MEDICATION UTILISATION DURING FLOOD DISASTERS:

A COMPARISON OF XGBOOST, LSTM AND TEMPORAL FUSION TRANSFORMER



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INTRODUCTION

- Flood disasters can **disrupt** healthcare services and affect medication demand.¹
- Accurate demand forecasting can **support medication planning** during disaster periods.²
- However, evidence on machine learning approaches for forecasting medication utilisation during floods **remains limited**.^{2,3}

OBJECTIVE

To compare the forecasting performance of

- 1) **XGBoost**,
 - 2) **Long Short-Term Memory (LSTM)**,
 - 3) **Temporal Fusion Transformer (TFT)**
- models for predicting cardiovascular medication utilisation across 7-, 14-, 21-, and 28-day forecast horizons.

METHODS

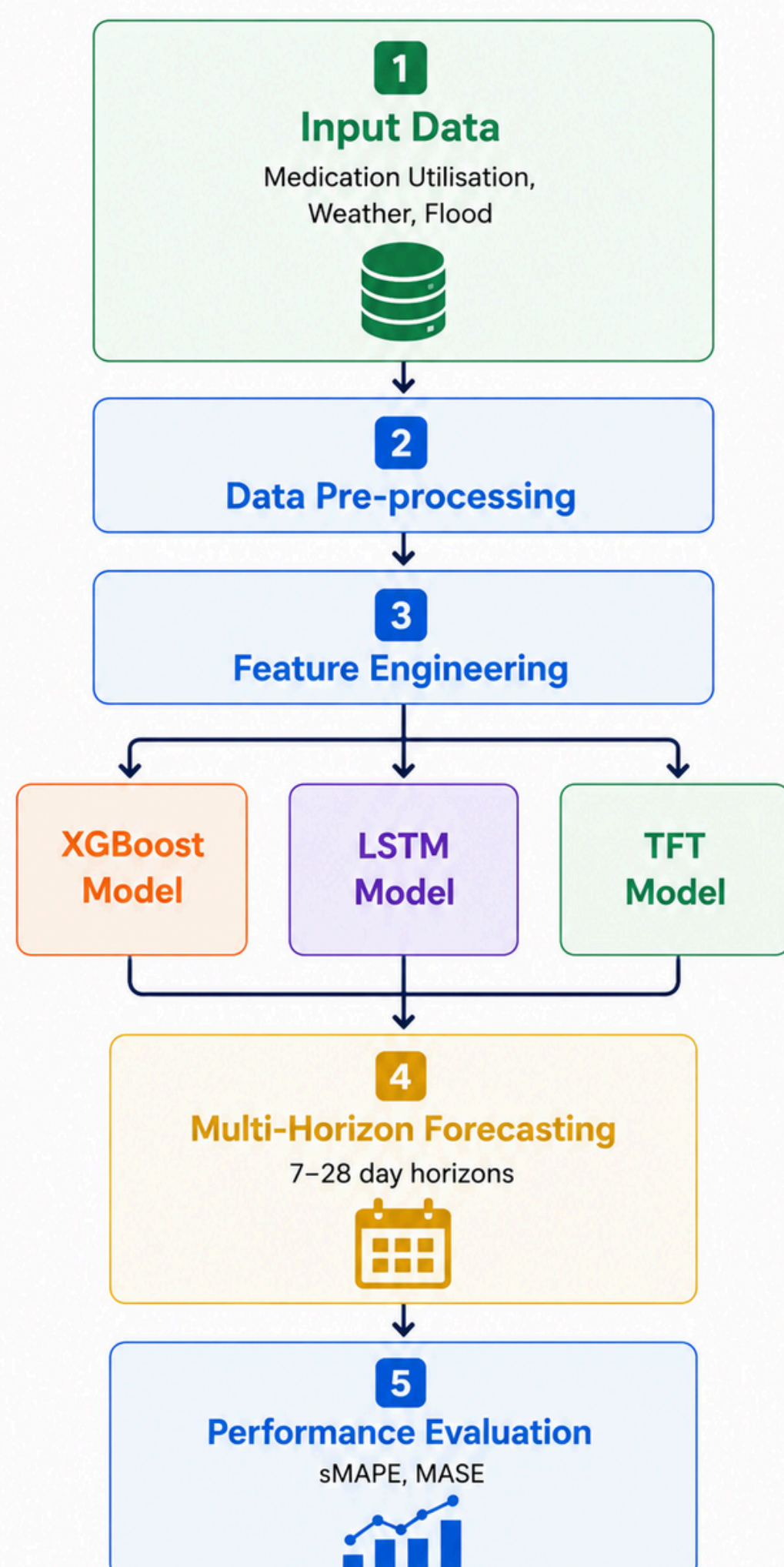


Figure 1. Overview of the proposed framework.

RESULTS & DISCUSSION

1 How does forecasting performance change across forecast horizons?

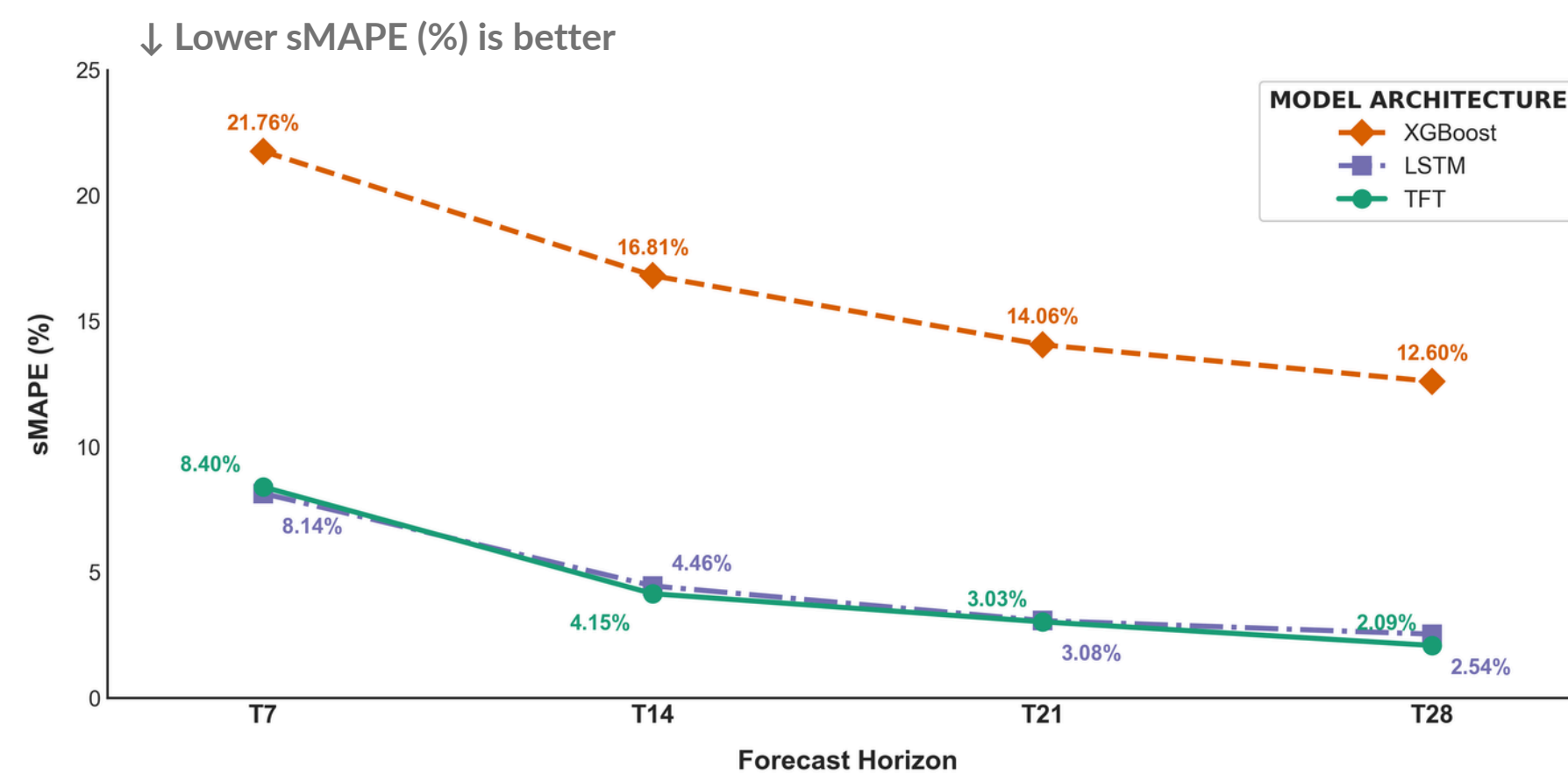


Figure 2. Overall forecasting performance across four horizons.

Prediction error **decreased** with longer forecast horizons.

This may be partly explained by the rolling-sum target. Longer forecast horizons aggregate medication utilisation over more days and smooth short-term variability. Therefore, lower error at longer forecast horizons does not necessarily mean that longer-term forecasting is easier.

2 How do models perform during flood observations?

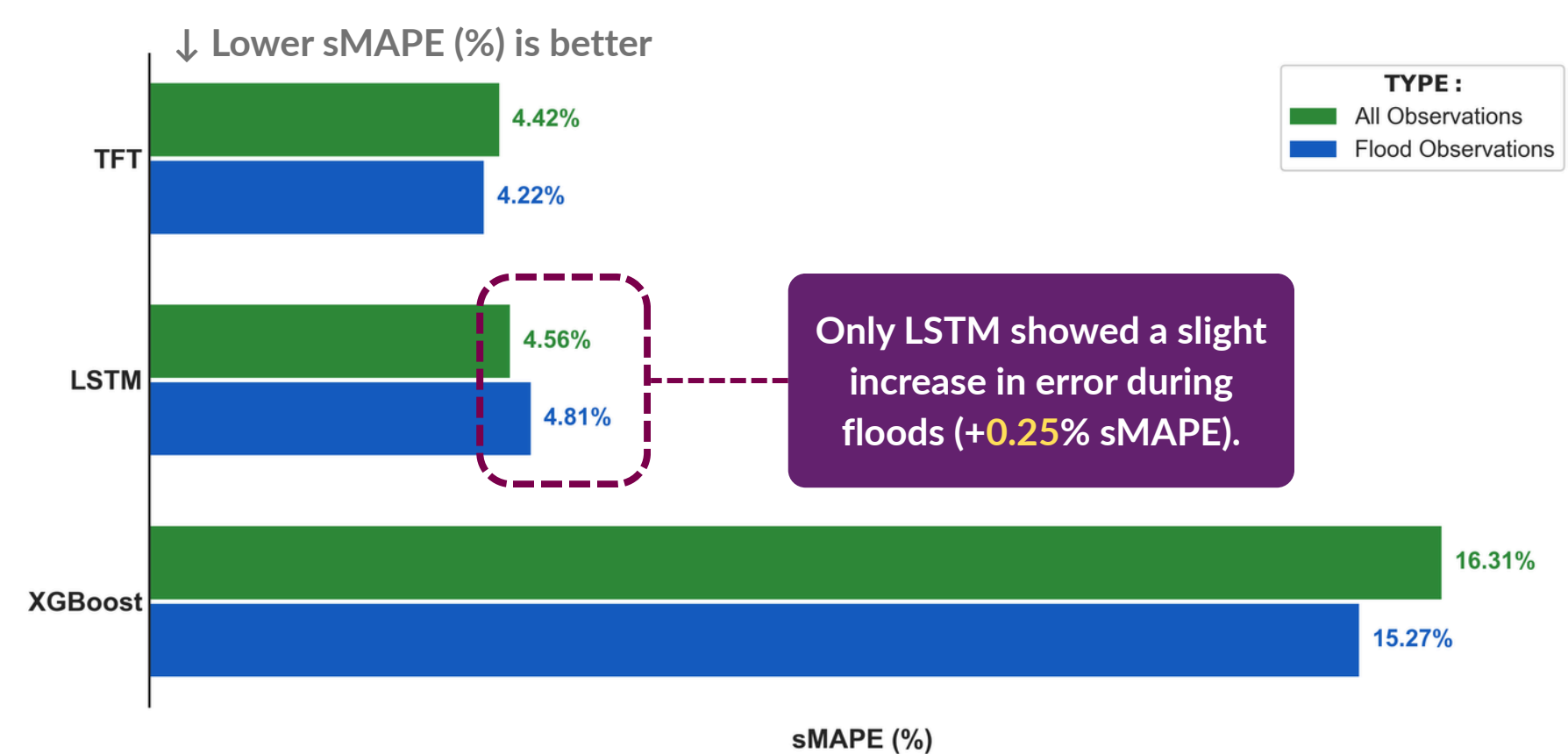


Figure 3. Mean sMAPE across 7- to 28-day horizons.

Model performance remained **stable** during flood observations, with only LSTM showing a slight increase in average sMAPE.

XGBoost showed higher error and relied heavily on historical utilisation features. LSTM and TFT incorporated a broader range of historical, operational and environmental information, which may have contributed to their stronger performance during flood observations.

3 Which model performs best across the tested settings?

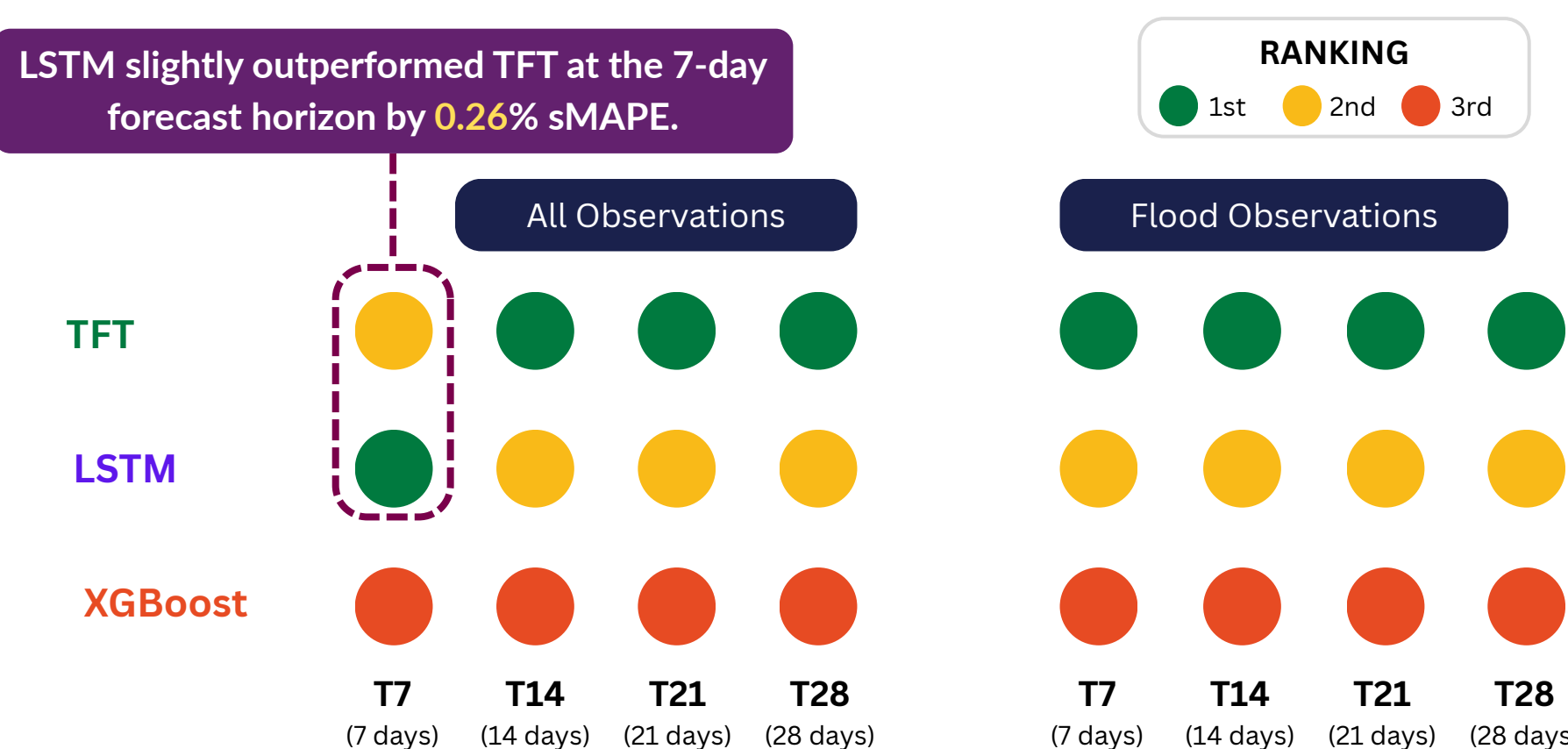


Figure 4. Model ranking across four horizons for overall and flood evaluations.

TFT was the best-performing model in 7 of 8 evaluation settings.

TFT variable selection retained historical utilisation together with temporal and flood-related variables, while temporal attention placed greater weight on recent observations. This may have contributed to TFT performance across forecast horizons.

CONCLUSION

Cardiovascular medication utilisation could be forecast up to **28 days**, with LSTM and TFT outperforming XGBoost. Forecasting error decreased at longer horizons, potentially reflecting smoothing from the rolling-sum targets. **TFT was the best-performing model** in 7 of 8 evaluation settings and may support medication planning during flood disasters.

FUTURE WORK

External validation using an **applicability domain** framework to evaluate model performance in new settings.

REFERENCES

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ACKNOWLEDGEMENT



MORE INFO

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